Droplet Paired-Tag v2.0

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Disclaimer

An automated process was used to import the protocol onto **protocols.io**. Please email yangxie@nygenome.org with the DOI of the protocol and any errors that you identify as we can fix these if necessary.

Abstract

This protocol is an updated version of “Droplet-based Single-cell Joint Profiling of Histone Modification and Transcriptome v1.” (https://www.protocols.io/view/droplet-based-single-cell-joint-profiling-of-histo-eq2ly6kergx9/v1)

In this version, we optimized multiple steps of the workflow and introduced indexed Tn5 adapters to enable sample multiplexing on the DNA level. Compared to original protocol, this updated version shows at least 3-5x increases in histone modification signal without compromising signal-to-noise ratio.

This updated protocol was used for generating high-quality single-cell multiomic data for BRAIN Initiative studies, including Chang et al., 2026 (PMID: 41684930) and Xie et al., 2026 (PMID: 41676679). We confirm that it accommodate both NextGEM and GEM-X platform from 10X Genomics.

Attachments

XLSX

PreIndex_adapters.xl...

9KB

Guidelines

For quality control and data analysis, please refer to https://github.com/Xieeeee/BICAN_DPT

Materials

nuclei lysis buffer (OMNI buffer)
ATAC-RSB buffer
Digitonin (2%)
IGEPAL-CA630 (10%)
Tween-20 (10%)
SUPERaseIN
IN RNaseOUT
50X PI
high salt Med Buffer #1 (MED1)
HEPES (pH 7.5, 1 M)
NaCl (5 M)
Spermidine (0.1 M)
RNaseOUT
EDTA (500 mM)
10% BSA
H_2O
antibody against specific histone modification epitope
assembled pA-Tn5 (0.4 mg/ml)
sorted cells
PBS
NeuN+ and NeuN- nuclei
0.5mL maximum recovery tubes
Med Buffer #2 (MED2)
$MgCl_2$
1X Nuclei buffer
20X Nuclei Buffer (10X Genomics PN-2000207)
DTT (100mM)
ThermoMixer
2X Stop Solution
cell counter
PCR tube
2xMaster Mix
N7XX primer
N5XX primer
Qiagen MinElute kit
0.5 mL tube
1xPBS
Buffer PB
MinElute column
2 ml collection tube
Buffer PE
enthanol
1.5 mL (or 2 mL) LoBind tube

Purified DNA
10 uM Nextera i5 primer (From Lei -20C box)
10 uM Nextera i7 primer (From Yang -20C box)
NEB 2xQ5 Master Mix
tapestation
tapestation buffer
10X Genomics Single Cell Multiome ATAC + GEX Gel Beads
barcoding reagents
ATAC Buffer B (10X Genomics PN-2000193)
Barcoding Reagent Mix (10X Genomics PN-2000267)
Template Switch Oligo (10X Genomics PN-3000228)
Reducing Agent B (10X Genomics PN-2000087)
Barcoding Enzyme Mix (10X Genomics PN-2000266)
Chromium Next GEM Chip J
ChromiumX microfluidics system (10X Genomics)
Bio-Rad T100 thermal cycler
Quenching Agent (10X Genomics PN-2000269)

ProteinA-Tn5 assembly

1 We synthesized all custom oligonucleotides at a concentration of 100 µM. Custom oligos were mixed in two separate PCR tubes (10 µL indexed AdapterA + 10 µL bridge-pMENTs, and 10 µL AdapterB + 10 µL pMENTs-B) and annealed in a thermal cycler using the following program:
95 °C for 5 min, then slow cooling to 12 °C, at 0.1 °C/s.
The 20 µL annealed oligos were mixed with 120 µL unloaded pA-Tn5 (0.5 mg/mL, ordered from MacroLab) in each tube and incubated at room temperature for 30 mins, followed by an additional 10 mins at 4 °C.

Finally, equal volumes of assembled pA-Tn5-indexed AdapterA/bridge-pMENTs and pA-Tn5-AdapterB/pMENTs-B were then combined to form functional indexed pA-Tn5 complexes, which were stored at -20 °C.

Freshly prepare the following buffer:

2 **A. Day 1: nuclei preparation and antibody staining**
A-1 Freshly prepare the following buffer:
Prepare nuclei lysis buffer (OMNI buffer) as: **250ul/sample*3 reactions**
693.75 µL ATAC-RSB buffer
3.75 µL Digitonin (2%)
7.5 µL IGEPAL-CA630 (10%)
7.5 µL Tween-20 (10%)
15 µL SUPERaseIN
7.5 µL IN RNaseOUT
15 µL 50X PI

3 Prepare the high salt Med Buffer #1 (MED1) as: **200ul/sample*3 reactions**
15 µL HEPES (pH 7.5, 1 M)
40.6 µL NaCl (5 M)
3.75 µL Spermidine (0.1 M)
15 µL 50X PI
18.75 µL SUPERaseIN
9.375 µL RNaseOUT
0.75 µL Digitonin (2%)
3.75 µL EDTA (500 mM)
150 µL 10% BSA
550 µL H_2O

Antibody + pA-Tn5 conjugation:

4 Mix antibody against specific histone modification epitope (2 µg) with 2µL assembled pA-Tn5 (0.4 mg/ml) in 30 µL MED1 buffer. Pellet mixing 20 times. No vortex! Be careful about liquid crosstalk between tubes.

	A	B	C	D
1	Tube#	Histone mark	Lot#	PA-Tn5
2	1	H3K27ac	A23	#1
3	2	H3K27me3	B22	#2
4	3	H3K9me3	C22	#3

Indexed pA-Tn5 usage is optional. If desired, indexed adapter sequences can be found in the provided supplementary table.

5 Rotate 1 hour at room temperature for antibody-pA-Tn5 conjugation.

6 Add 45ul MED#1 buffer to each antibody-pA-Tn5 conjugation tube.

Nuclei preparation (for NeuN sorted samples):

7 Spin down sorted cells at 300 RCF for 5 mins and then 500 RCF for 10 mins. Leave less than 200ul buffer per tube and combine the same type together.

8 Cell counting (1:4 dilute PBS) and combine NeuN+ and NeuN- nuclei together (1:1) to more than 500k/sample. Aliquot nuclei to 0.5mL maximum recovery tubes.

	A	B	C	D	E	F
1	Sample Name	NeuN+ Con.	NeuN+ Volume	NeuN- Con.	NeuN- Volume	Total (Ratio)
2						
3						
4						

9 Spin down cells at 300RCF 5 mins+500 RCF for 5 mins. Resuspend the cells in 240 µL OMNI buffer. Sit on ice for 3mins 30s. (Start timer after adding OMNI buffer to the 1st tube)

10 Spin down nuclei at 700 RCF for 5 mins at 4°C and resuspend nuclei in 120 µL MED1 Buffer.

Epitope targeting:

11 Spin down the nuclei at 300 RCF for 10 min at 4°C and resuspended in 75 µL Buffer (check volume) with conjugated antibody-pA-Tn5 to each tube. Incubate the mixture with rotation at 4 °C overnight.

Freshly prepare the following buffer:

12 **B. Day2: multi-omics tagging and nuclei barcoding**
B-1 Freshly prepare the following buffer:
Prepare the Med Buffer #2 (MED2) as: **250ul/sample*3 reactions**
15 µL HEPES (pH 7.5, 1 M)
40.6 µL NaCl (5 M)
3.75 µL spermidine (0.1 M)
15 µL 50X PI
18.75 µL SUPERaseIN
9.375 µL RNaseOUT
0.75 µL digitonin (2%)
150 µL 10% BSA
553 µL H_2O
(MED2 buffer + $MgCl_2$): 75 µL MED2 buffer + 0.75 µL 1M $MgCl_2$ per sample
Aliquot 225 µL MED2 buffer + 2.25 µL 1M $MgCl_2$

13 Prepare 1X Nuclei buffer as: **200ul/sample*3 reactions+100ul extra**
35 µL 20X Nuclei Buffer (10X Genomics PN-2000207) 5 µL
7 µL DTT (100mM) 1 µL
17.5 µL SUPERaseIN 2.5 µL
8.75 µL RNaseOUT 1.25 µL
631.75 µL H2O 90.25 µL

Nuclei washing and tagmentation:

14 Spin down nuclei at 300 RCF for 10 mins at 4°C, washed with 75 µL MED2 buffer. Repeat washing for twice.

15 Resuspend washed nuclei in 70 µL (MED2 buffer + $MgCl_2$). Rotate in ThermoMixer at 550 r.p.m., 37 °C, for 1 hr.

16 Terminate tagmentation by adding 70 µL 2X Stop Solution, mixing well. Spin down nuclei at 500 RCF for 6 mins at 4°C, wash once with 100 µL 1X Nuclei buffer.

17 Spin down nuclei at 500 RCF for 6 mins at 4°C, and resuspend in 100 µL 1X Nuclei buffer. Spin down nuclei at 500 RCF for 6 mins at 4°C, and remove 90ul supernatant. Count

nuclei using cell counter (1:10 dilute PBS), transfer all the nuclei into PCR tube on ice, and wait for QC results.

	A	B	C	D
1	Tube number	Cell con.	Aliquot	1x Nuclei buffer
2				
3				

QC:

18 Thaw the reagents for PCR (including 2xMaster Mix, Nextera primers (N7XX and N5XX)

19 Purify tagmented DNA with Qiagen MinElute kit from remaining nuclei in the 0.5 mL tube, and elute with 21 µL H2O.

19.1 Add 100 µL 1xPBS to each 0.5 mL tube.

19.2 Add 5 volumes of Buffer PB to 1 volume of the sample and vortex. (Add 500 µL of Buffer PB to 100 µL 1xPBS.)

19.3 Place a MinElute column in a provided 2 ml collection tube in a suitable rack.

19.4 To bind DNA, apply the sample to the MinElute column, wait for 1 min, and centrifuge at 13,000xg for 1 min.

19.5 Discard flow-through. Place the MinElute column back into the same tube.

19.6 To wash, add 700 µL Buffer PE (Check if enthanol is added to Buffer PE) to the MinElute column and centrifuge for 1 min.

19.7 Discard flow-through and place the purple column back in the same collection tube.

19.8 Centrifuge the column for an additional 3 min.

19.9 Discard the collection tube, place the MinElute column in a 1.5 mL (or 2 mL) LoBind tube. Open the cap to air dry for 1 min.

19.10 Add 21ul of H2O to the membrane, bind for 1 min.

19.11 Centrifuge the column for an additional 1 min.

20 Assemble the indexing PCR mix as below:
20 uL Purified DNA
2.5 uL 10 uM Nextera i5 primer (From Lei -20C box)
2.5 uL 10 uM Nextera i7 primer (From Yang -20C box)
25 uL NEB 2xQ5 Master Mix

21 PCR program:
72C 5 min
98C 30 sec
98C 10 sec *
63C 30 sec * Repeat for a total of (1+12) cycles
72C 1 min *
72C 2 min
4C Hold

22 Take 2ul from PCR reaction and add 8ul H2O. Then, aliquot 2 ul for tapestation+2 ul tapestation buffer.

23 During this time, thaw 10X Genomics Single Cell Multiome ATAC + GEX Gel Beads at and barcoding reagents at room temperature.

Droplet generation, reverse transcription and cell barcoding in droplet:

24 Combine nuclei according to the QC result:
Per tube:
+ x uL Nuclei (20k using larger nuclei concerntration)
+ (8-x) uL 1x Nuclei Buffer
+ 7 µL ATAC Buffer B (10X Genomics PN-2000193)

25 Prepare reverse transcription and cell barcoding master mix as:

	A	B
1	1.1x N reactions	1x
2	µL Barcoding Reagent Mix (10X Genomics PN-2000267)	49.5 µL

	A	B
3	µL Template Switch Oligo (10X Genomics PN-3000228)	1.1 µL
4	µL Reducing Agent B (10X Genomics PN-2000087)	1.9 µL
5	µL Barcoding Enzyme Mix (10X Genomics PN-2000266)	7.5 µL

26 Add 60 µL prepared master mix to 15 µL nuclei mixture, loaded onto Chromium Next GEM Chip J and proceed to droplet generation with ChromiumX microfluidics system (10X Genomics). Detailed processes of chip assembly and reagents adding are the same as described in 10X Multiome user guide (10X Genomics CG000338).

27 After droplet generation completed, aspirate 100 µl emulsions (nuclei-encapsulating droplets) into a new PCR tube. Incubated in a Bio-Rad T100 thermal cycler with following program, with lid temperature set to 50 °C: (Reverse transcription as well as cell barcoding complete during the incubation.)
37 °C 45mins
25 °C 30mins hold at 4 °C

28 After incubation, add 5 µl Quenching Agent (10X Genomics PN-2000269) to stop the reaction. Slowly pipette mixing well, and stored at -80C.